

Fermi-liquid behavior of nonaltermagnetic RuO₂Maxim Wenzel,¹ Ece Uykur^{2,*}, Sahana Rößler³, Marcus Schmidt,³ Oleg Janson⁴, Achyut Tiwari,¹ Martin Dressel¹, and Alexander A. Tsirlin^{5,†}¹*Physikalisches Institut, Universität Stuttgart, D-70569 Stuttgart, Germany*²*Helmholtz-Zentrum Dresden-Rossendorf, Institute of Ion Beam Physics and Materials Research, D-01328 Dresden, Germany*³*Max Planck Institute for Chemical Physics of Solids, D-01067 Dresden, Germany*⁴*Institute for Theoretical Solid State Physics, Leibniz IFW Dresden, D-01069 Dresden, Germany*⁵*Felix Bloch Institute for Solid-State Physics, University of Leipzig, D-04103 Leipzig, Germany*

(Received 15 July 2024; revised 22 October 2024; accepted 8 January 2025; published 28 January 2025)

The presence of magnetism in potentially altermagnetic RuO₂ has been a subject of intense debate. Using broadband infrared spectroscopy combined with density-functional band-structure calculations, we show that the optical conductivity of RuO₂, the bulk probe of its electronic structure, is best described by a nonmagnetic model. The sharp Pauli edge demonstrates the presence of a Dirac nodal line lying 45 meV below the Fermi level. An excellent match between the experimental and *ab initio* plasma frequencies underpins the weakness of electronic correlations. The intraband part of the optical conductivity indicates Fermi-liquid behavior with two distinct scattering rates below 150 K. Fermi-liquid theory also accounts for the temperature-dependent magnetic susceptibility of RuO₂ and allows a consistent description of this material as a paramagnetic metal.

DOI: [10.1103/PhysRevB.111.L041115](https://doi.org/10.1103/PhysRevB.111.L041115)

Introduction. Ruthenium dioxide is a simple inorganic solid that finds applications in catalysis [1], microelectronics [2], and supercapacitors [3]. The recent comeback of interest in this material has been triggered by the proposed realization of altermagnetism, owing to the nonsymmorphic nature of the RuO₂ crystal structure [4–9]. Yet, the bare presence of ordered magnetic moments in this material is disputed. Resonant x-ray scattering [10,11] and neutron diffraction experiments [12] pinpointed ordered magnetic moments on the Ru atoms. Moreover, angle-resolved photoemission spectroscopy (ARPES) performed using circularly polarized soft x-rays demonstrated the splitting of spin-polarized bands expected in altermagnets [13]. On the other hand, magneto-transport measurements that allow a bulk probe of the Fermi surface [14] are consistent with the nonmagnetic band structure [15–17]. Moreover, recent muon spin relaxation [18] and renewed neutron diffraction [19,20], and ARPES experiments [21] do not find indications of bulk magnetism in RuO₂, thus leaving magnetic instability and, consequently, realization of altermagnetism in this material highly controversial.

Some of the key applications of altermagnets hinge upon their unusual transport properties. Here, we study electron dynamics in RuO₂ using optical spectroscopy that probes the bulk electronic structure of this material across a broad range of energies, both above and below the Fermi level.

Surprisingly, previous optical measurements mainly focused on the *p-d* transitions above 2 eV [22–26], thus leaving unexplored the most interesting energy range covering Ru valence bands and their itinerant carriers. Using broadband infrared spectroscopy combined with density-functional calculations, we demonstrate that the optical response of RuO₂ is well described by the nonmagnetic and uncorrelated band structure. On the other hand, the typical spin-split band structure of an altermagnet fails to reproduce the essential features of the experimental spectra, most notably, the Dirac nodal line manifested by the Pauli edge [27] in the optical conductivity. This nodal line lies only 45 meV below the Fermi level and represents the most prominent feature in the valence electronic states of bulk RuO₂. By comparing the experimental and calculated plasma frequencies, we show that RuO₂ should be classified as a weakly correlated paramagnetic metal. The intraband part of its optical response manifests Fermi-liquid behavior, which is also consistent with the weak and continuous increase in the magnetic susceptibility with temperature.

Methods. Single crystals of RuO₂ have been grown by chemical vapor transport [28] and characterized by magnetic susceptibility as well as dc-electrical resistivity measurements [inset of Fig. 1(b)]. Apart from the Curie tail at low temperatures, magnetic susceptibility weakly increases on heating up to at least 400 K, consistent with the previous reports [12,36–38], whereas electrical transport shows the typical metallic behavior with the residual resistivity ratio (RRR) of 118, which is on par with the best samples used in transport measurements of bulk crystals [39–42] and almost two orders of magnitude higher than the typical RRR values of the thin-film samples [5,8,43–46].

Reflectivity measurements were performed with the Bruker Vertex80v and IFS113v spectrometers in a broad frequency range (50–20 000 cm^{−1}) on as-grown single crystals

*Contact author: e.uykur@hzdr.de†Contact author: altsirlin@gmail.com

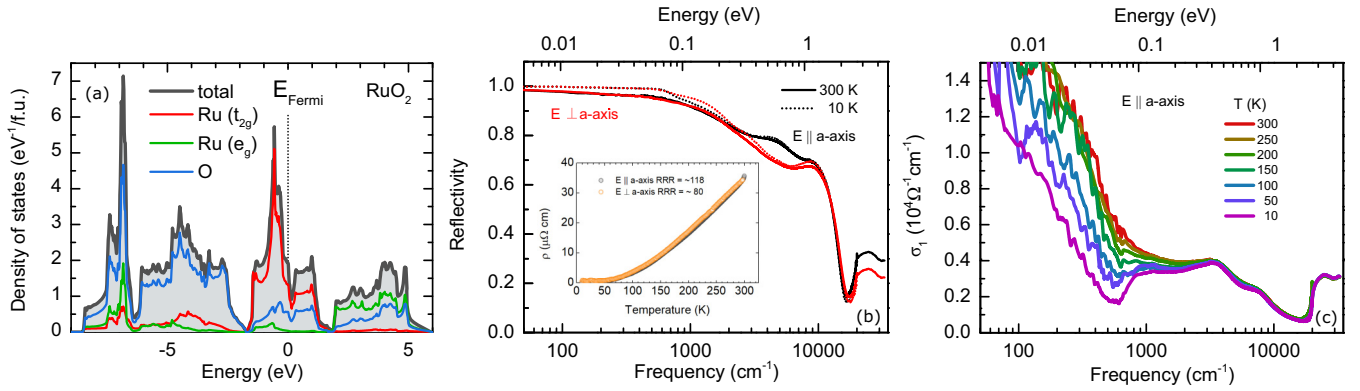


FIG. 1. (a) Density of states for the nonmagnetic RuO_2 . The states near the Fermi energy are dominated by the Ru t_{2g} orbitals. (b) Broadband reflectivity measured along and perpendicular to the a axis at room temperature and 10 K. No temperature dependence is observed, except at low energies where the intraband transitions are dominant. (c) Temperature-dependent in-plane optical conductivity. The inset in (b) shows the dc resistivity of our RuO_2 single crystal.

[Fig. 1(b)]. The (011) surface was used for polarization-dependent measurements, thus allowing two nonequivalent directions, $E \parallel a$ axis and $E \perp a$ axis (mostly $E \parallel c$ [28]). Optical conductivity was obtained by the Kramers-Kronig transformation of the reflectivity data. We used Hagen-Rubens extrapolations in the low-energy regime, whereas the high-energy region was extrapolated by using the ellipsometry data collected at room temperature up to 6 eV (unpolarized) and x-ray scattering functions above 6 eV [47]. The temperature-dependent spectra for the in-plane conductivity are shown in Fig. 1(c). The data for $E \perp a$ can be found in the Supplemental Material [28].

Density-functional band-structure calculations were performed on the full-relativistic level in the WIEN2K [48,49] and full-potential local-orbital (FPLO) [50] codes using the experimental crystal structure from Ref. [51] and Perdew-Burke-Ernzerhof (PBE) exchange-correlation potential [52]. The nonmagnetic band structure was obtained from PBE + SO (spin-orbit) calculations. Additionally, the mean-field density-functional theory (DFT)+ U correction was introduced for stabilizing the magnetic solution similar to the previous studies [53–57]. Optical conductivity was obtained by the internal routine of WIEN2K [58].

Band structure and optical conductivity. Figure 1(a) shows the density of states for nonmagnetic RuO_2 . The states near the Fermi level are dominated by the Ru t_{2g} orbitals and separated by small gaps from the Ru e_g states above 2 eV and from the O $2p$ states below -2 eV. These separations are clearly visible in the optical response of the material. Indeed, calculated optical conductivity of the nonmagnetic RuO_2 [Fig. 2(b)] shows a pronounced minimum around 2 eV that separates optical transitions within the t_{2g} bands from the O $2p \rightarrow$ Ru t_{2g} and Ru $t_{2g} \rightarrow$ Ru e_g transitions.

Experimental interband optical conductivity obtained by subtracting the intraband response reveals a relatively weak anisotropy and clearly shows the minimum at 1.5–2.0 eV separating the transitions within the t_{2g} manifold from the higher-energy transitions. Three main absorption features shown by arrows in Fig. 2(a) appear below this minimum. All of them can be traced in the calculated optical conductivity obtained from the nonmagnetic band structure. This

nonmagnetic band structure correctly shows the crossing of the $E \parallel a$ and $E \perp a$ conductivities around 0.65 eV, as well as their merger in the energy range of the minimum, around 2.0 eV.

The magnetic solution is unstable in PBE + SO and requires the addition of electronic correlations that change the

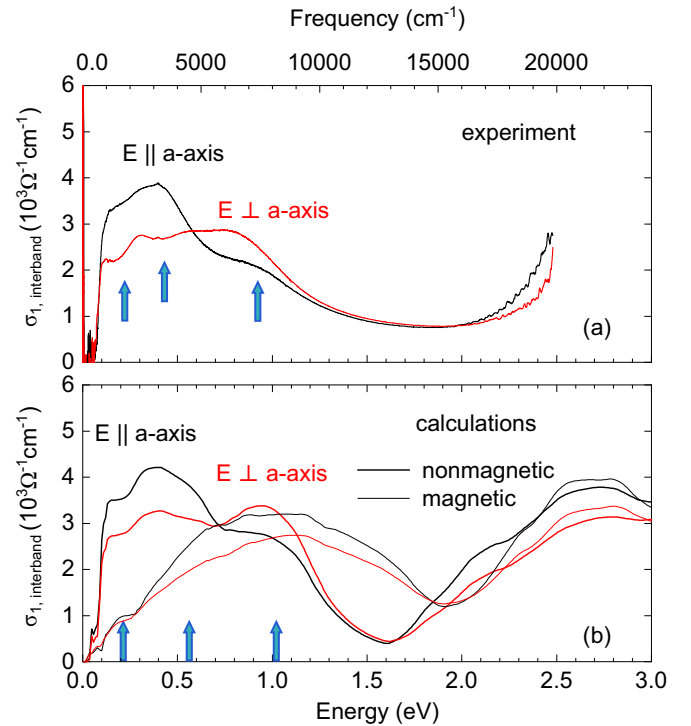


FIG. 2. (a) Experimental interband optical conductivity of RuO_2 for $E \parallel a$ and $E \perp a$ obtained by subtracting the intraband contribution from the spectra. The main absorption features are shown with arrows. (b) Calculated interband optical conductivity for the nonmagnetic band structure of RuO_2 (PBE + SO) and for the altermagnetic solution ($U = 2$ eV, $J = 0.3$ eV). Frequency-dependent broadening (up to 0.1 eV at 1 eV) is applied to the calculated spectra for an easier comparison with the experiment. The energy axis for the calculated spectra is rescaled by the factor of 1.3 to account for a renormalization of the band energies.

positions of the Ru bands and shift the spectral weight to higher energies. By adding correlations we obtain a solution with a sizable spin splitting of about 1 eV that corresponds to the typical band structure of the putative RuO₂ altermagnet [4,5]. The minimum in the optical conductivity is noticeably shifted to higher energies and becomes less pronounced [Fig. 2(b)]. Adding magnetism clearly deteriorates the description of the experimental optical conductivity. Note in particular the acute discrepancies at low energies where the sharp increase in the optical conductivity (Pauli edge) cannot be reproduced by the magnetic solution.

The best match between the experimental spectrum and the optical conductivity obtained from the nonmagnetic solution is achieved when the calculated energies are divided by the factor of 1.3. Such rescaling is not uncommon in weakly correlated electronic systems where band energies are slightly renormalized compared to the band picture [59]. The magnitude of electronic correlations can be gauged by comparing the experimental and calculated plasma frequencies or the respective intraband spectral weights [60–62]. Our nonmagnetic calculation returns $\omega_p^{xx} = 3.29$ eV and $\omega_p^{zz} = 3.45$ eV in a very good agreement with the experimental values of 3.16 and 3.34 eV for $E \parallel a$ and $E \perp a$, respectively. The ratio of the spectral weights of the intraband contribution, $SW_{\text{exp}}/SW_{\text{DFT}} = 0.87 \pm 0.05$, further identifies RuO₂ as a weakly correlated metal. This value is comparable to materials such as CsV₃Sb₅ [61] and ZrSiS [63] where correlation effects play only a minor role.

The magnetic solution returns $\omega_p^{xx} = 2.65$ eV and $\omega_p^{zz} = 2.58$ eV and clearly underestimates the plasma frequencies of RuO₂. We thus conclude that bulk RuO₂ can be best described on the level of standard band theory without including magnetism or the on-site Coulomb repulsion U .

Ru t_{2g} bands. Previous optical studies of RuO₂ did not probe the energy range below 0.5–1.0 eV and could not fully resolve the optical transitions within the t_{2g} bands [22,64,65]. With the data extending to low energies, we identify three main features in this region (Fig. 3). The sharp increase in the optical conductivity around 0.1 eV followed by the almost flat region up to 0.4 eV is reminiscent of the Pauli edge in Dirac semimetals [27,66]. At higher energies, σ_{xx} peaks at 0.5–0.6 eV and develops a visible shoulder around 1.0 eV.

Our band-resolved calculations show that the spectral weight below 0.4 eV mostly originates from the bands *B* and *C* that form a Dirac nodal line, which ends at the gapped Dirac point along the *A-M* direction. The sharp Pauli edge observed in the interband optical conductivity around 0.1 eV is a direct experimental fingerprint of this topological feature [53,67] that should lead to the large spin-Hall effect in RuO₂ [68]. Other possible low-energy transitions, for example between the linear bands *A* and *B* along *Z-Γ*, give only a small contribution in this energy range. Above 0.4 eV, optical transitions along the *X-R* line start contributing to the spectrum, resulting in the peak of the optical conductivity. The shoulder around 1.0 eV is due to multiple interband transitions and reflects the fact that several bands develop almost flat regions at -0.5 and $+0.5$ eV, respectively.

The interband part of the experimental spectrum is decomposed using the Pauli edge and three Lorentzian features [Fig. 3(c)] that show a striking agreement with the individual

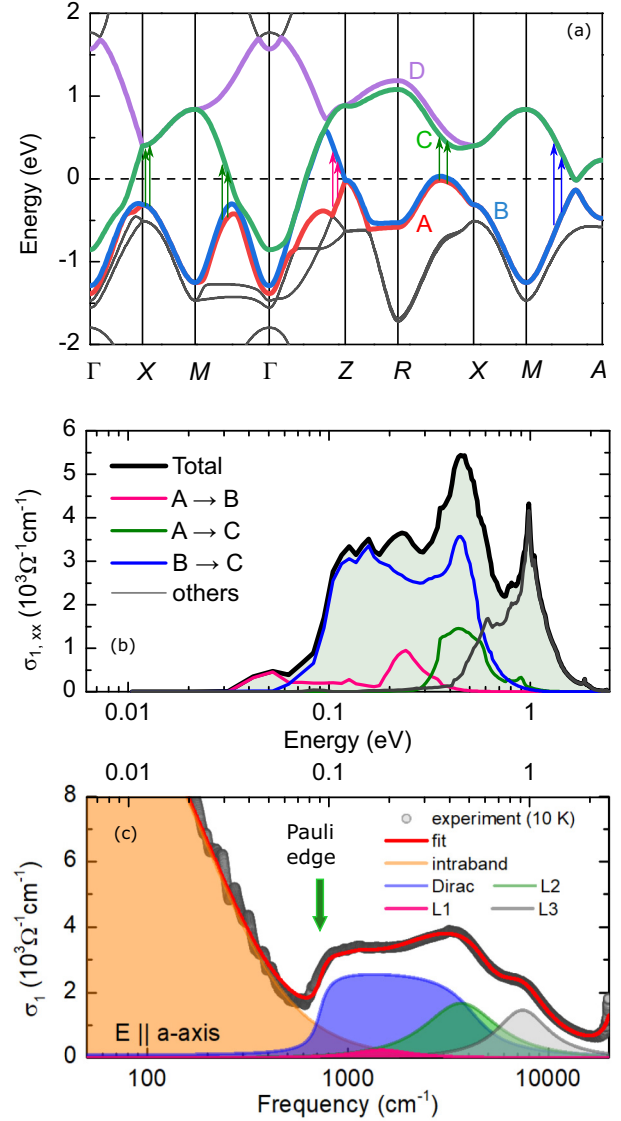


FIG. 3. (a) Band structure of nonmagnetic RuO₂. Arrows indicate the major absorption features, which are color coded for the band-resolved optical conductivity. (b) Band-resolved optical conductivity with the energy axis divided by the factor of 1.3. No broadening is included into the calculation. Three major absorption features are shown with colored lines, whereas the absorption around 1 eV shown with the gray line is the combination of several interband transitions. (c) Experimental optical conductivity at 10 K and the decomposition of the spectrum into the intraband contribution and the interband contribution that consists of the Pauli edge and several Lorentzian peaks [28] showing a remarkable agreement with the band-resolved optical conductivity from DFT.

contributions from the band-resolved analysis. Our experimental data put the Pauli edge to 90 meV and pin the Dirac nodal line of bulk RuO₂ to 45 meV below the Fermi level. It is worth noting that adding magnetism eliminates this Dirac nodal line [28] and leads to an optical conductivity without the prominent Pauli edge [Fig. 2(b)]. Therefore, the observation of the Pauli edge serves as direct experimental evidence for the nonmagnetic solution as the best starting point for addressing the electronic structure of bulk RuO₂.

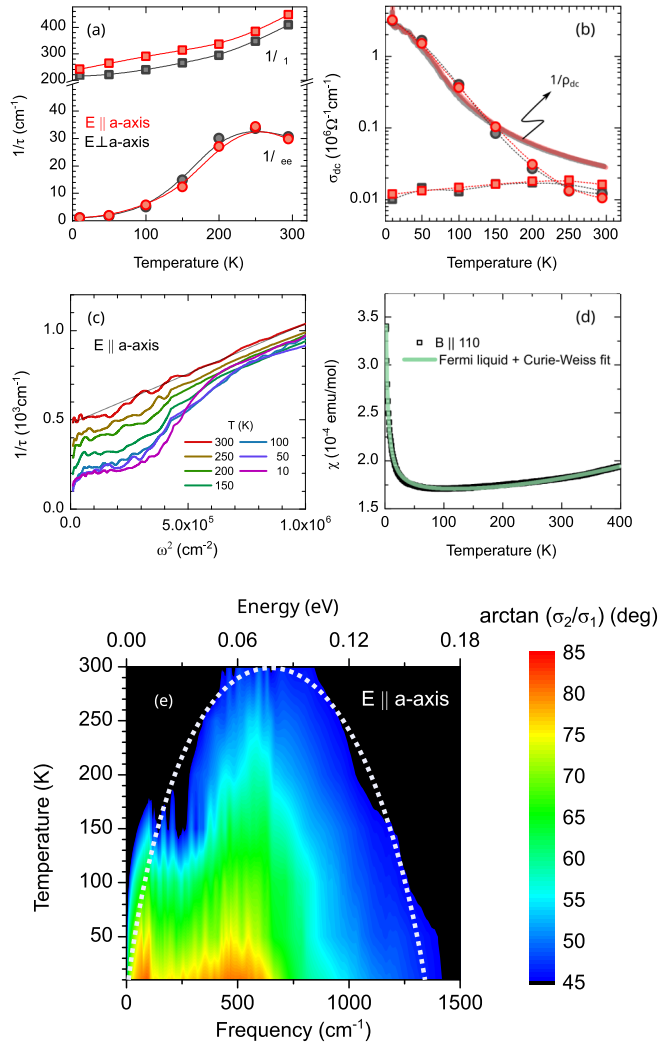


FIG. 4. (a) Temperature-dependent scattering rates of the intraband contribution and the corresponding dc conductivities overlaid with the values obtained from the measured dc resistivity (b). (c) Quadratic frequency dependence of the scattering rate of the extended Drude model. (d) Temperature-dependent magnetic susceptibility of RuO₂ measured in the applied field of 1 T and its fit using the T^2 term of the paramagnetic metal along with the Curie-like impurity contribution [28]. (e) The phase angle σ_2/σ_1 determined from our optical experiments in the T - ω domain. The semielliptical region of $\sigma_2 > \sigma_1$ witnesses the Fermi-liquid behavior.

Intraband contribution. Having elucidated the interband part of the optical conductivity, we now discuss the optical response of itinerant carriers. Interestingly, this response cannot be modeled with the conventional Drude peak. Instead, our data reveal the presence of two distinct scattering rates that both increase on heating [Fig. 4(a)]. The respective contributions to the dc conductivity shown in Fig. 4(b) suggest that at low temperatures the dc conductivity of RuO₂ is mainly determined by the lower relaxation rate, whereas above 200 K the contributions of both relaxation rates are comparable. This situation is reminiscent of an interplay between phonon scattering and electron-electron scattering, with the former becoming predominant at elevated temperatures. A similar scenario was also described theoretically for a

system with two bands and the interband electron-electron scattering [69].

Alternatively, the intraband part of the optical conductivity can be analyzed using the extended Drude model with the frequency-dependent relaxation rate, $1/\tau(\omega)$ [60,70]. This analysis reveals the quadratic frequency dependence typical of a Fermi liquid [Fig. 4(c)]. Below 150 K, the difference between the two scattering rates becomes prominent, and a step in the frequency dependence appears. The quadratic behavior is nevertheless preserved at frequencies both above and below the step. The Fermi-liquid behavior is further witnessed by the phase angle of the imaginary and real parts of the optical conductivity, σ_2/σ_1 , plotted in the T - ω domain in Fig. 4(e). The semielliptical shape of the region where $\sigma_2 > \sigma_1$ is typical of the Fermi-liquid regime [71,72].

Discussion. Our results suggest that the bulk electronic structure of RuO₂ can be well understood on the level of band theory without including correlations or magnetism. The analysis of the intraband spectral weight gives direct evidence for the weakness of electronic correlations in this material. It further supports the nonmagnetic nature of bulk RuO₂ because the addition of the on-site Coulomb repulsion is essential for stabilizing magnetic moments on the Ru atoms. Adding magnetism deteriorates the agreement with the experiment for both intraband and interband parts of the spectrum.

The possible altermagnetism of RuO₂ has been ascribed to a Fermi-surface instability that mostly arises from the small pockets of the Fermi surface along X - R and A - M [12,53]. Minor changes in the position of the Fermi level, e.g., by a variation of the oxygen content, should significantly affect these parts of the Fermi surface and may thus change the proclivity of RuO₂ for magnetism. This renders the oxygen nonstoichiometry scenario [55] a viable explanation for the presence of magnetism in thin films and absence of magnetism in the bulk, especially in light of the drastically low RRR values reported for the thin-film samples [5,8,43–46]. Optical spectroscopy would be a useful tool for probing the true electronic structure of such thin films. In particular, the position of the Pauli edge can serve as a sensitive probe of the Fermi level.

In the absence of magnetism, bulk RuO₂ should be thought of as a paramagnetic metal. Indeed, the intraband part of its optical conductivity reveals the Fermi-liquid type of behavior with the quadratic scaling of the scattering rate. This finding justifies the application of Fermi-liquid theory also to the magnetic properties of bulk RuO₂. Temperature-dependent susceptibilities of paramagnetic transition metals have been previously described using the T^2 term that arises in Fermi-liquid theory [73–75] as well as in band theory [76,77]. Figure 4(d) shows that a similar description is possible in the case of RuO₂ too when the low-temperature upturn is taken into account by adding the Curie-like contribution that corresponds to only 0.17% of spin- $\frac{1}{2}$ impurities [28]. The zero-temperature Pauli susceptibility of RuO₂ is $\chi_0 = 1.63 \times 10^{-4}$ emu/mol, in very good agreement with 1.9×10^{-4} emu/mol obtained in DFT for the nonmagnetic solution. This comparison further verifies the Fermi-liquid description of the magnetic susceptibility. The monotonic increase in the magnetic susceptibility of RuO₂ is not an indication of antiferromagnetism. It manifests the paramagnetic response of itinerant

carriers, similar to simple $4d$ transition metals such as Ru and Rh [78].

Our optical data not only support the absence of magnetism in bulk RuO_2 , as inferred from various probes of magnetic order [18,19], but also give strong evidence for the weakness of electronic correlations in this material. These data set an important constraint on stabilizing magnetism within the RuO_2 framework. The apparent signatures of altermagnetism reported in thin-film samples [5–9,13] suggest that effects such as strain and off-stoichiometry may render RuO_2 magnetic. However, this putative magnetism has to be assessed without invoking sizable electronic correlations, contrary to the existing theoretical studies that typically enforce magnetism within a correlated scenario.

Conclusions. The electrodynamics of RuO_2 probed by the broadband optical spectroscopy is well described by the nonmagnetic band structure of this material. The interband part of the optical conductivity reveals several absorption peaks due to optical transitions within the Ru t_{2g} bands along with the Pauli edge that manifests the Dirac nodal line located

45 meV below the Fermi level. The analysis of the plasma frequencies and intraband spectral weight classifies RuO_2 as a weakly correlated metal. Its intraband optical response abides by the Fermi-liquid theory, which also explains the monotonic increase in the magnetic susceptibility and offers a consistent description of bulk RuO_2 as a paramagnetic metal. Taken together, these results suggest that it is not necessary to invoke (alter)magnetism for describing the electronic properties of bulk ruthenium dioxide.

Acknowledgments. We are grateful to Vicky Hasse for technical assistance with crystal growth and Gabriele Untereiner for preparing single crystals for optical measurements. M.W. is supported by Center for Integrated Quantum Science and Technology (IQST) Stuttgart/Ulm via a project funded by the Carl Zeiss Stiftung. The work in HZDR is funded by the DFG UY 63/5-1. The work in Leipzig was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – TRR 360 – 492547816 (subproject B1).

Data availability. Figure data in Originlab format and computational details of the nodal lines can be found in Ref. [79].

-
- [1] H. Over, Surface chemistry of ruthenium dioxide in heterogeneous catalysis and electrocatalysis: From fundamental to applied research, *Chem. Rev.* **112**, 3356 (2012).
 - [2] G. S. Iles and M. E. A. Casale, Ruthenium oxide glaze resistors, *Platinum Metal Rev.* **11**, 126 (1967).
 - [3] D. Majumdar, T. Maiyalagan, and Z. Jiang, Recent progress in ruthenium oxide-based composites for supercapacitor applications, *ChemElectroChem* **6**, 4343 (2019).
 - [4] L. Šmejkal, J. Sinova, and T. Jungwirth, Emerging research landscape of altermagnetism, *Phys. Rev. X* **12**, 040501 (2022).
 - [5] Z. Feng, X. Zhou, L. Šmejkal, L. Wu, Z. Zhu, H. Guo, R. González-Hernández, X. Wang, H. Yan, P. Qin, X. Zhang, H. Wu, H. Chen, Z. Meng, L. Liu, Z. Xia, J. Sinova, T. Jungwirth, and Z. Liu, An anomalous Hall effect in altermagnetic ruthenium dioxide, *Nat. Electron.* **5**, 735 (2022).
 - [6] A. Bose, N. J. Schreiber, R. Jain, D.-F. Shao, H. P. Nair, J. Sun, X. S. Zhang, D. A. Muller, E. Y. Tsymlal, D. G. Schlom, and D. C. Ralph, Tilted spin current generated by the collinear antiferromagnet ruthenium dioxide, *Nat. Electron.* **5**, 267 (2022).
 - [7] H. Bai, L. Han, X. Y. Feng, Y. J. Zhou, R. X. Su, Q. Wang, L. Y. Liao, W. X. Zhu, X. Z. Chen, F. Pan, X. L. Fan, and C. Song, Observation of spin splitting torque in a collinear antiferromagnet RuO_2 , *Phys. Rev. Lett.* **128**, 197202 (2022).
 - [8] S. Karube, T. Tanaka, D. Sugawara, N. Kadoguchi, M. Kohda, and J. Nitta, Observation of spin-splitting torque in collinear antiferromagnetic RuO_2 , *Phys. Rev. Lett.* **129**, 137201 (2022).
 - [9] S. G. Jeong, I. H. Choi, S. Nair, L. Buiarelli, B. Pourbahari, J. Y. Oh, N. Bassim, A. Seo, W. S. Choi, R. M. Fernandes, T. Birol, L. Zhao, J. S. Lee, and B. Jalan, Altermagnetic polar metallic phase in ultra-thin epitaxially-strained RuO_2 films, [arXiv:2405.05838](https://arxiv.org/abs/2405.05838).
 - [10] Z. H. Zhu, J. Strempler, R. R. Rao, C. A. Occhialini, J. Pellicciari, Y. Choi, T. Kawaguchi, H. You, J. F. Mitchell, Y. Shao-Horn, and R. Comin, Anomalous antiferromagnetism in metallic RuO_2 determined by resonant x-ray scattering, *Phys. Rev. Lett.* **122**, 017202 (2019).
 - [11] B. Z. Gregory, J. Strempler, D. Weinstock, J. P. Ruf, Y. Sun, H. Nair, N. J. Schreiber, D. G. Schlom, K. M. Shen, and A. Singer, Strain-induced orbital-energy shift in antiferromagnetic RuO_2 revealed by resonant elastic x-ray scattering, *Phys. Rev. B* **106**, 195135 (2022).
 - [12] T. Berlijn, P. C. Snijders, O. Delaire, H.-D. Zhou, T. A. Maier, H.-B. Cao, S.-X. Chi, M. Matsuda, Y. Wang, M. R. Koehler, P. R. C. Kent, and H. H. Weitering, Itinerant antiferromagnetism in RuO_2 , *Phys. Rev. Lett.* **118**, 077201 (2017).
 - [13] O. Fedchenko, J. Minár, A. Akashdeep, S. W. D'Souza, D. Vasilyev, O. Tkach, L. Odenbreit, Q. Nguyen, D. Kutnyakhov, N. Wind, L. Wenthaus, M. Scholz, K. Rossnagel, M. Hoesch, M. Aeschlimann, B. Stadtmüller, M. Kläui, G. Schönhense, T. Jungwirth, A. B. Hellenes *et al.*, Observation of time-reversal symmetry breaking in the band structure of altermagnetic RuO_2 , *Sci. Adv.* **10**, eadj4883 (2024).
 - [14] J. E. Graebner, E. S. Greiner, and W. D. Ryden, Magnetothermal oscillations in RuO_2 , OsO_2 , and IrO_2 , *Phys. Rev. B* **13**, 2426 (1976).
 - [15] L. F. Mattheiss, Electronic structure of RuO_2 , OsO_2 , and IrO_2 , *Phys. Rev. B* **13**, 2433 (1976).
 - [16] B. Y. Yavorsky, O. V. Krasovska, E. E. Krasovskii, A. N. Yaresko, and V. N. Antonov, *Ab initio* calculation of the Fermi surface of RuO_2 , *Phys. B: Condens. Matter* **225**, 243 (1996).
 - [17] X. Peng, Z. Liu, S. Zhang, Y. Zhou, Y. Sun, Y. Su, C. Wu, T. Zhou, L. Liu, Y. Li, H. Wang, J. Yang, B. Chen, Y. Li, C. Xi, J. Du, Z. Jiao, Q. Wu, and M. Fang, Universal scaling behavior of transport properties in non-magnetic RuO_2 , [arXiv:2412.12258](https://arxiv.org/abs/2412.12258).
 - [18] M. Hiraishi, H. Okabe, A. Koda, R. Kadono, T. Muroi, D. Hirai, and Z. Hiroi, Nonmagnetic ground state in RuO_2 revealed by muon spin rotation, *Phys. Rev. Lett.* **132**, 166702 (2024).
 - [19] P. Keßler, L. Garcia-Gassull, A. Suter, T. Prokscha, Z. Salman, D. Khalyavin, P. Manuel, F. Orlandi, I. I. Mazin, R. Valentí, and S. Moser, Absence of magnetic order in RuO_2 : insights from

- μ SR spectroscopy and neutron diffraction, *npj Spintronics* **2**, 50 (2024).
- [20] L. Kiefer, F. Wirth, A. Bertin, P. Becker, L. Bohatý, K. Schmalzl, A. Stunault, J. A. Rodríguez-Velamazán, O. Fabelo, and M. Braden, Crystal structure and absence of magnetic order in single crystalline RuO₂, [arXiv:2410.05850](https://arxiv.org/abs/2410.05850).
- [21] J. Liu, J. Zhan, T. Li, J. Liu, S. Cheng, Y. Shi, L. Deng, M. Zhang, C. Li, J. Ding, Q. Jiang, M. Ye, Z. Liu, Z. Jiang, S. Wang, Q. Li, Y. Xie, Y. Wang, S. Qiao, J. Wen *et al.*, Absence of altermagnetic spin splitting character in rutile oxide RuO₂, *Phys. Rev. Lett.* **133**, 176401 (2024).
- [22] A. K. Goel, G. Skorinko, and F. H. Pollak, Optical properties of single-crystal rutile RuO₂ and IrO₂ in the range 0.5 to 9.5 eV, *Phys. Rev. B* **24**, 7342 (1981).
- [23] H. L. Park, C. H. Chung, C. H. Kim, and H. S. Kim, Optical properties of RuO₂ thin film, *J. Mater. Sci. Lett.* **6**, 1093 (1987).
- [24] A. Belkind, Z. Orban, J. L. Vossen, and J. A. Woollam, Optical properties of RuO₂ films deposited by reactive sputtering, *Thin Solid Films* **207**, 242 (1992).
- [25] P. Hones, T. Gerfin, and M. Grätzel, Spectroscopic ellipsometry of RuO₂ films prepared by metalorganic chemical vapor deposition, *Appl. Phys. Lett.* **67**, 3078 (1995).
- [26] G. Mondio, F. Neri, M. Allegrini, A. Iembo, and F. Fuso, Energy loss spectroscopy of RuO₂ thin films, *J. Appl. Phys.* **82**, 1730 (1997).
- [27] A. V. Pronin and M. Dressel, Nodal semimetals: A survey on optical conductivity, *Phys. Status Solidi B* **258**, 2000027 (2021).
- [28] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/PhysRevB.111.L041115> for details of the sample growth and characterization as well as auxiliary DFT results, additional optical data, and their analysis, which includes Refs. [29–35].
- [29] K. Momma and F. Izumi, VESTA 3 for three-dimensional visualization of crystal, volumetric and morphology data, *J. Appl. Crystallogr.* **44**, 1272 (2011).
- [30] J. P. Carbotte, Optical response of a line node semimetal, *J. Phys.: Condens. Matter* **29**, 045301 (2017).
- [31] S. P. Mukherjee and J. P. Carbotte, Transport and optics at the node in a nodal loop semimetal, *Phys. Rev. B* **95**, 214203 (2017).
- [32] S. Ahn, E. J. Mele, and H. Min, Electrodynamics on Fermi cyclides in nodal line semimetals, *Phys. Rev. Lett.* **119**, 147402 (2017).
- [33] S. J. Youn, T. H. Rho, B. I. Min, and K. S. Kim, Extended Drude model analysis of noble metals, *Phys. Status Solidi B* **244**, 1354 (2007).
- [34] A. M. Awasthi, L. Degiorgi, G. Grüner, Y. Dalichaouch, and M. B. Maple, Complete optical spectrum of CeAl₃, *Phys. Rev. B* **48**, 10692 (1993).
- [35] G. Bossé, L. S. Bilbro, R. V. Aguilar, L. D. Pan, W. Liu, A. V. Stier, Y. Li, L. H. Greene, J. Eckstein, and N. P. Armitage, Low energy electrodynamics of the Kondo-lattice antiferromagnet CeCu₂Ge₂, *Phys. Rev. B* **85**, 155105 (2012).
- [36] A. N. Guthrie and L. T. Bourland, Magnetic susceptibilities and ionic moments in the palladium and platinum groups, *Phys. Rev.* **37**, 303 (1931).
- [37] J. M. Fletcher, W. E. Gardner, B. F. Greenfield, M. J. Holdoway, and M. H. Rand, Magnetic and other studies of ruthenium dioxide and its hydrate, *J. Chem. Soc. A*, 653 (1968).
- [38] W. D. Ryden and A. W. Lawson, Magnetic susceptibility of IrO₂ and RuO₂, *J. Chem. Phys.* **52**, 6058 (1970).
- [39] W. D. Ryden, A. W. Lawson, and C. C. Sartain, Electrical transport properties of IrO₂ and RuO₂, *Phys. Rev. B* **1**, 1494 (1970).
- [40] Y. S. Huang, H. L. Park, and F. H. Pollak, Growth and characterization of RuO₂ single crystals, *Mater. Res. Bull.* **17**, 1305 (1982).
- [41] J. J. Lin, S. M. Huang, Y. H. Lin, T. C. Lee, H. Liu, X. X. Zhang, R. S. Chen, and Y. S. Huang, Low temperature electrical transport properties of RuO₂ and IrO₂ single crystals, *J. Phys.: Condens. Matter* **16**, 8035 (2004).
- [42] F. Pawula, A. Fakih, R. Daou, S. Hébert, N. Mordvinova, O. Lebedev, D. Pelloquin, and A. Maignan, Multiband transport in RuO₂, *Phys. Rev. B* **110**, 064432 (2024).
- [43] J. J. Lin, W. Xu, Y. L. Zhong, J. H. Huang, and Y. S. Huang, Electron-electron scattering times in low-diffusivity thick RuO₂ and IrO₂ films, *Phys. Rev. B* **59**, 344 (1999).
- [44] M. Uchida, T. Nomoto, M. Musashi, R. Arita, and M. Kawasaki, Superconductivity in uniquely strained RuO₂ films, *Phys. Rev. Lett.* **125**, 147001 (2020).
- [45] J. P. Ruf, H. Paik, N. J. Schreiber, H. P. Nair, L. Miao, J. K. Kawasaki, J. N. Nelson, B. D. Faeth, Y. Lee, B. H. Goodge, B. Pamuk, C. J. Fennie, L. F. Kourkoutis, D. G. Schlom, and K. M. Shen, Strain-stabilized superconductivity, *Nat. Commun.* **12**, 59 (2021).
- [46] J. Liu, L. Gao, Y. T. Zou, T. Lin, M. T. Zhu, X. Y. Lyu, C. Lu, Y. Q. Wang, A. L. Ji, Q. H. Zhang, Z. G. Cheng, L. Gu, Z. X. Cao, and N. P. Lu, Emergent weak antilocalization and wide-temperature-range electronic phase diagram in epitaxial RuO₂ thin film, *J. Phys.: Condens. Matter* **35**, 405603 (2023).
- [47] D. B. Tanner, Use of x-ray scattering functions in Kramers-Kronig analysis of reflectance, *Phys. Rev. B* **91**, 035123 (2015).
- [48] P. Blaha, K. Schwarz, G. Madsen, D. Kvasnicka, J. Luitz, R. Laskowski, F. Tran, and L. Marks, *WIEN2k, An Augmented Plane Wave + Local Orbitals Program for Calculating Crystal Properties* (Karlheinz Schwarz, Technische Universität Wien, Austria, 2018).
- [49] P. Blaha, K. Schwarz, F. Tran, R. Laskowski, G. K. H. Madsen, and L. D. Marks, WIEN2k: An APW+lo program for calculating the properties of solids, *J. Chem. Phys.* **152**, 074101 (2020).
- [50] K. Koepnick and H. Eschrig, Full-potential nonorthogonal local-orbital minimum-basis band-structure scheme, *Phys. Rev. B* **59**, 1743 (1999).
- [51] J. Haines, J. M. Léger, O. Schulte, and S. Hull, Neutron diffraction study of the ambient-pressure, rutile-type and the high-pressure, CaCl₂-type phases of ruthenium dioxide, *Acta Crystallogr., Sect. B* **53**, 880 (1997).
- [52] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77**, 3865 (1996).
- [53] K.-H. Ahn, A. Hariki, K.-W. Lee, and J. Kuneš, Antiferromagnetism in RuO₂ as *d*-wave Pomeranchuk instability, *Phys. Rev. B* **99**, 184432 (2019).
- [54] L. Šmejkal, A. Marmodoro, K.-H. Ahn, R. González-Hernández, I. Turek, S. Mankovsky, H. Ebert, S. W. D'Souza, O. Šipr, J. Sinova, and T. Jungwirth, Chiral magnons in altermagnetic RuO₂, *Phys. Rev. Lett.* **131**, 256703 (2023).
- [55] A. Smolyanyuk, I. I. Mazin, L. García-Gassull, and R. Valentí, Fragility of the magnetic order in the prototypical altermagnet RuO₂, *Phys. Rev. B* **109**, 134424 (2024).

- [56] X. Zhou, W. Feng, R.-W. Zhang, L. Šmejkal, J. Sinova, Y. Mokrousov, and Y. Yao, Crystal thermal transport in altermagnetic RuO₂, *Phys. Rev. Lett.* **132**, 056701 (2024).
- [57] A. Hariki, Y. Takahashi, and J. Kuneš, X-ray magnetic circular dichroism in RuO₂, *Phys. Rev. B* **109**, 094413 (2024).
- [58] C. Ambrosch-Draxl and J. Sofo, Linear optical properties of solids within the full-potential linearized augmented planewave method, *Comput. Phys. Commun.* **175**, 1 (2006).
- [59] E. Uykur, B. R. Ortiz, S. D. Wilson, M. Dressel, and A. A. Tsirlin, Optical detection of the density-wave instability in the kagome metal KV₃Sb₅, *npj Quantum Mater.* **7**, 16 (2022).
- [60] D. N. Basov, R. D. Averitt, D. van der Marel, M. Dressel, and K. Haule, Electrodynamics of correlated electron materials, *Rev. Mod. Phys.* **83**, 471 (2011).
- [61] M. Wenzel, B. R. Ortiz, S. D. Wilson, M. Dressel, A. A. Tsirlin, and E. Uykur, Optical study of RbV₃Sb₅: Multiple density-wave gaps and phonon anomalies, *Phys. Rev. B* **105**, 245123 (2022).
- [62] M. Wenzel, E. Uykur, A. A. Tsirlin, S. Pal, R. M. Roy, C. Yi, C. Shekhar, C. Felser, A. V. Pronin, and M. Dressel, Intriguing low-temperature phase in the antiferromagnetic kagome metal FeGe, *Phys. Rev. Lett.* **132**, 266505 (2024).
- [63] Y. Shao, A. N. Rudenko, J. Hu, Z. Sun, Y. Zhu, S. Moon, A. J. Millis, S. Yuan, A. I. Lichtenstein, D. Smirnov, Z. Q. Mao, M. I. Katsnelson, and D. N. Basov, Electronic correlations in nodal-line semimetals, *Nat. Phys.* **16**, 636 (2020).
- [64] O. V. Krasovska, E. E. Krasovskii, and V. N. Antonov, *Ab initio* calculation of the optical and photoelectron properties of RuO₂, *Phys. Rev. B* **52**, 11825 (1995).
- [65] J. S. de Almeida and R. Ahuja, Electronic and optical properties of RuO₂ and IrO₂, *Phys. Rev. B* **73**, 165102 (2006).
- [66] M. B. Schilling, L. M. Schoop, B. V. Lotsch, M. Dressel, and A. V. Pronin, Flat optical conductivity in ZrSiS due to two-dimensional Dirac bands, *Phys. Rev. Lett.* **119**, 187401 (2017).
- [67] V. Jovic, R. J. Koch, S. K. Panda, H. Berger, P. Bugnon, A. Magrez, K. E. Smith, S. Biermann, C. Jozwiak, A. Bostwick, E. Rotenberg, and S. Moser, Dirac nodal lines and flat-band surface state in the functional oxide RuO₂, *Phys. Rev. B* **98**, 241101(R) (2018).
- [68] Y. Sun, Y. Zhang, C.-X. Liu, C. Felser, and B. Yan, Dirac nodal lines and induced spin Hall effect in metallic rutile oxides, *Phys. Rev. B* **95**, 235104 (2017).
- [69] D. L. Maslov and A. V. Chubukov, Optical response of correlated electron systems, *Rep. Prog. Phys.* **80**, 026503 (2017).
- [70] C. Berthod, J. Mravlje, X. Deng, R. Žitko, D. van der Marel, and A. Georges, Non-Drude universal scaling laws for the optical response of local Fermi liquids, *Phys. Rev. B* **87**, 115109 (2013).
- [71] A. Tytarenko, Y. Huang, A. de Visser, S. Johnston, and E. van Heumen, Direct observation of a Fermi liquid-like normal state in an iron-pnictide superconductor, *Sci. Rep.* **5**, 12421 (2015).
- [72] A. Pustogow, Y. Saito, A. Löhle, M. S. Alonso, A. Kawamoto, V. Dobrosavljević, M. Dressel, and S. Fratini, Rise and fall of Landau's quasiparticles while approaching the Mott transition, *Nat. Commun.* **12**, 1571 (2021).
- [73] S. Misawa and K. Kanematsu, Susceptibility maximum and Fermi-liquid effect in 4*d* and 5*d* transition metals, *J. Phys. F* **6**, 2119 (1976).
- [74] G. M. Carneiro and C. J. Pethick, Finite-temperature contributions to the magnetic susceptibility of a normal Fermi liquid, *Phys. Rev. B* **16**, 1933 (1977).
- [75] A. V. Chubukov and D. L. Maslov, Nonanalytic corrections to the Fermi-liquid behavior, *Phys. Rev. B* **68**, 155113 (2003).
- [76] C. J. Kiessman and H. B. Callen, The magnetic susceptibility of the transition elements, *Phys. Rev.* **94**, 837 (1954).
- [77] M. Shimizu, Itinerant electron magnetism, *Rep. Prog. Phys.* **44**, 329 (1981).
- [78] E. V. Galoshina, Magnetic susceptibility of *d*-band transition metals that are not magnetically ordered, *Sov. Phys. Usp.* **17**, 345 (1974).
- [79] M. Wenzel, E. Uykur, S. Rößler, M. Schmidt, O. Janson, A. Tiwari, M. Dressel, and A. A. Tsirlin, Data set: Fermi-liquid behavior of non-altermagnetic RuO₂, Rodare (2025), doi: 10.14278/rodare.3542.